

Aeronautical Systems: Sessions 4-

1. Linearized Equations of Motion
2. Dimensional Stability Derivatives and Non-Dimensional Stability Derivatives

Equilibrium Conditions

Consider steady, straight, and level flight without sideslip. Quantities at the equilibrium point are denoted by subscript 0.

- Velocity : $U_0, V_0 = 0, W_0$
- Angular Velocity : $P_0 = Q_0 = R_0 = 0$
- : $\Phi_0 = \phi_0, \Theta_0 = \theta_0, \Psi_0 = \psi_0 = 0$ (Set the yaw angle of the equilibrium state to 0.)
- External Force : $X_{a0}, Y_{a0}, Z_{a0}, L_0 = 0, M_0 = 0, N_0 = 0$

Small Perturbation

Equilibrium Table.

- Velocity : $U = U_0 + u, V = v, W = W_0 + w$ (A2-1a)
- Angular Velocity : $P = p, Q = q, R = r$ (A2-1b)
- Euler angle : $\Phi = \phi_0 + \phi, \Theta = \theta_0 + \theta, \Psi = \psi$ (A2-1c)

Changes from the equilibrium point are denoted by Δ .

- External Force (Aerodynamic force) : $X_a = X_{a0} + \Delta X_a, Y_a = Y_{a0} + \Delta Y_a, Z_a = Z_{a0} + \Delta Z_a$ (A2-1d)
- External Force (Aerodynamic moment) : $L = \Delta L, M = \Delta M, N = \Delta N$ (A2-1e)

Equilibrium Equation

At the equilibrium condition, the left-hand sides of Eqs. (A1-47) and (A1-48) become zero, yielding:

$$X_{a0} - mg \sin \theta_0 = 0 \quad (\text{A2-2a})$$

$$Y_{a0} + mg \cos \theta_0 \sin \phi_0 = 0 \quad (\text{A2-2b})$$

$$Z_{a0} + mg \cos \theta_0 \cos \phi_0 = 0 \quad (\text{A2-2c})$$

$$L_0 = 0 \quad (\text{A2-2d})$$

$$M_0 = 0 \quad (\text{A2-2e})$$

$$N_0 = 0 \quad (\text{A2-2f})$$

- Velocity : $U_0, V_0 = 0, W_0$
- Angular Velocity : $P_0 = Q_0 = R_0 = 0$
- $\Phi_0 = \phi_0, \Theta_0 = \theta_0, \Psi_0 = \psi_0 = 0$
- External Force : $X_{a0}, Y_{a0}, Z_{a0}, L_0 = 0, M_0 = 0, N_0 = 0$

$$m(\dot{U} + QW - RV) = -mg \sin \Theta + X_a \quad (\text{A1-47a})$$

$$m(\dot{V} + RU - PW) = mg \cos \Theta \sin \Phi + Y_a \quad (\text{A1-47b})$$

$$m(\dot{W} + PV - QU) = mg \cos \Theta \cos \Phi + Z_a \quad (\text{A1-47c})$$

$$I_{xx}\dot{P} - I_{xz}\dot{R} + (I_{zz} - I_{yy})QR - I_{xz}PQ = L \quad (\text{A1-48a})$$

$$I_{yy}\dot{Q} + (I_{xx} - I_{zz})RP + I_{xz}(P^2 - R^2) = M \quad (\text{A1-48b})$$

$$-I_{zx}\dot{P} + I_{zz}\dot{R} + (I_{yy} - I_{xx})PQ + I_{xz}QR = N \quad (\text{A1-48c})$$

Equations of Motion for Small Perturbations

By substituting Eq. (A2-1) into Eqs. (A1-47) and (A1-48), and considering Eq. (A2-2), the equations of motion for small perturbations are obtained.

Small Perturbation for Equation

$$m(\dot{u} + qW_0) = -\theta mg \cos \theta_0 + \Delta X_a \quad (\text{A2-3a})$$

$$m(\dot{v} + rU_0 - pW_0) = \phi mg \cos \theta_0 + \Delta Y_a \quad (\text{A2-3b})$$

$$m(\dot{w} - qU_0) = -\theta mg \sin \theta_0 + \Delta Z_a \quad (\text{A2-3c})$$

$$I_{xx}\dot{p} - I_{xz}\dot{r} = \Delta L \quad (\text{A2-3d})$$

$$I_{yy}\dot{q} = \Delta M \quad (\text{A2-3e})$$

$$-I_{zx}\dot{p} + I_{zz}\dot{r} = \Delta N \quad (\text{A2-3f})$$

Substitute Eq. (A2-1) into Eq. (A1-62):

Body Angular Velocity → Euler Angles

$$\dot{\phi} = p + r \tan \theta_0 \quad (\text{A2-4a})$$

$$\dot{\theta} = q \quad (\text{A2-4b})$$

$$\dot{\psi} = r \sec \theta_0 \quad (\text{A2-4c})$$

- Velocity : $U = U_0 + u, V = v, W = W_0 + w \quad (\text{A2-1a})$

- Angular Velocity : $P = p, Q = q, R = r \quad (\text{A2-1b})$

- Euler angle : $\Phi = \phi_0 + \phi, \Theta = \theta_0 + \theta, \Psi = \psi \quad (\text{A2-1c})$

$$m(\dot{U} + QW - RV) = -mg \sin \Theta + X_a \quad (\text{A1-47a})$$

$$m(\dot{V} + RU - PW) = mg \cos \Theta \sin \Phi + Y_a \quad (\text{A1-47b})$$

$$m(\dot{W} + PV - QU) = mg \cos \Theta \cos \Phi + Z_a \quad (\text{A1-47c})$$

$$I_{xx}\dot{P} - I_{xz}\dot{R} + (I_{zz} - I_{yy})QR - I_{xz}PQ = L \quad (\text{A1-48a})$$

$$I_{yy}\dot{Q} + (I_{xx} - I_{zz})RP + I_{xz}(P^2 - R^2) = M \quad (\text{A1-48b})$$

$$-I_{zx}\dot{P} + I_{zz}\dot{R} + (I_{yy} - I_{xx})PQ + I_{xz}QR = N \quad (\text{A1-48c})$$

November 20: Final Lecture - Derivation of Linearized Motion of Equations

December 27: **Class Cancelled** due to conference; Report Assignment Uploaded to LMS

Q4

- Modeling External Forces in Motion Equations
- Estimating Aerodynamic Forces on the Airframe
- Motion Analysis Using Linearized Motion Equations(MATLAB)